

A1.

(1) $3 \times 3 = 9$

(2) $(0, a), (1, a), (2, a)$

(3) $(0, a), (0, b), (0, c)$

A4.

(1) $ABUACUBC$

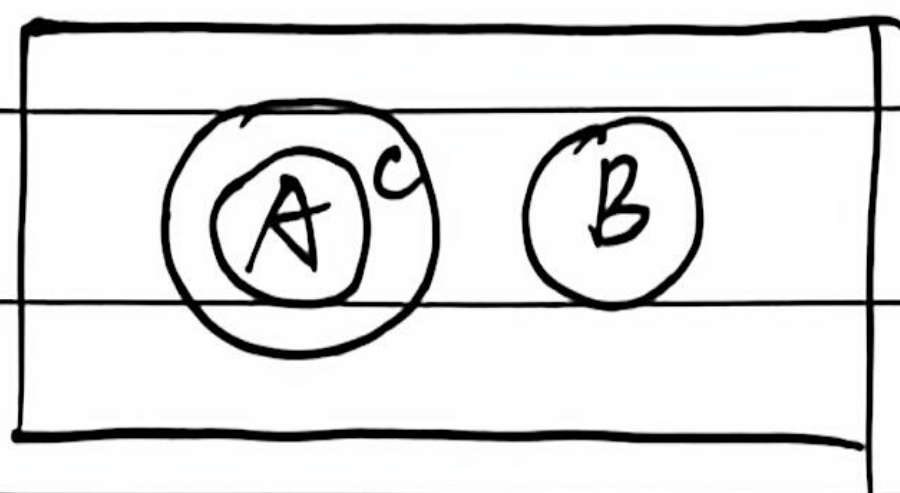
(2) $\overline{ABUACUBC}$

(3) $ABUACUBC \cap \overline{ABC}$

(4) $\overline{ABC}$

B1.

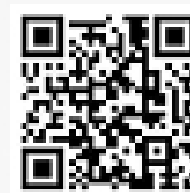
(1) $A \cap C, B \cap C = \emptyset$



$P_1 = P(C) - P(A) = 0.1$

(2) $P(A \cup B) = P(A) + P(B) = 0.6$

(3) $P(\overline{A \cup B \cup C}) = P(\overline{B \cup C}) = 1 - P(B) - P(C) = 0.3$



A12.

(i) 有放回抽样

设 A_i 为第 i 次摸到红球, $P(A_i) = \frac{8}{10} = \frac{4}{5}$, 且结果独立

$$(1) P(A_1 A_2) = \left(\frac{4}{5}\right)^2 = \frac{16}{25}$$

$$(2) P(A_1 \bar{A}_2 \cup \bar{A}_1 A_2) = \frac{4}{5} \times \frac{1}{5} + \frac{1}{5} \times \frac{4}{5} = \frac{8}{25}$$

$$(3) P(A_2) = \frac{4}{5}$$

(ii) 无放回抽样

$$(1) P(A_1 A_2) = \frac{C_8^2}{C_{10}^2} = \frac{28}{45}$$

$$(2) P(A_1 \bar{A}_2 \cup \bar{A}_1 A_2) = \frac{C_8^1 \cdot C_2^1}{C_{10}^2} = \frac{16}{45}$$

$$(3) P(A_2) = \frac{4}{5} \times \frac{7}{9} + \frac{1}{5} \times \frac{8}{9} = \frac{4}{5}$$

A14.

记 (1) 事件为 A, (2) 事件为 B, (3) 事件为 C

$$(1) P(A) = \frac{C_2^1 \cdot C_3^1 \cdot C_2^1}{C_7^3} = \frac{12}{35}$$

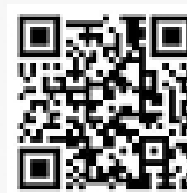
$$(2) P(B) = \frac{C_3^3 \cdot C_1^1}{C_7^3} = \frac{1}{35}$$

$$(3) P(C) = \frac{2}{7} \times \frac{3}{6} \times \frac{2}{5} = \frac{2}{35}$$

A15.

设 A_i 为第 i 号车配对.

$$(1) P(A_1) = \frac{8!}{9!} = \frac{1}{9}$$



CS 扫描全能王

3亿人都在用的扫描App

$$(2) P(A|\bar{A}_9) = \frac{7 \times A_7^7}{A_9^9} = \frac{7}{72}$$

A16.

设 A 为 3 号盒子中恰好有两个球

$$P(A) = \frac{C_5^2 \cdot 2^3}{3^5} = \frac{80}{243}$$

A19.

$$(1) P(A|A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A \cup B)} = \frac{P(A)}{P(A) + P(B) - P(AB)}$$

$$P(B) = 1 - P(\bar{B}) = 0.4, \quad P(AB) = P(A) - P(A\bar{B}) = 0.2$$

$$\Rightarrow P(A|A \cup B) = \frac{0.7}{0.9} = \frac{7}{9}$$

$$(2) P(A|\bar{A} \cup B) = \frac{P(A \cap (\bar{A} \cup B))}{P(\bar{A} \cup B)} = \frac{P(AB)}{P(\bar{A} \cup B)}$$

$$P(\bar{A}) = 1 - P(A) = 0.3, \quad P(\bar{A}B) = P(B) - P(AB) = 0.2$$

$$P(\bar{A} \cup B) = P(\bar{A}) + P(B) - P(\bar{A}B) = 0.5$$

$$P(A|\bar{A} \cup B) = \frac{2}{5}$$

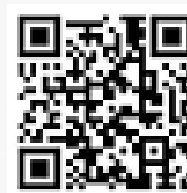
$$(3) P(AB|A \cup B) = \frac{P(AB \cap (A \cup B))}{P(A \cup B)} = \frac{P(AB)}{P(A \cup B)} = \frac{2}{9}$$

B4.

设 A 为女职工, B 为在管理岗位.

$$P(A) = 0.45, \quad P(B) = 0.1, \quad P(AB) = 0.05$$

$$(1) P(B|A) = \frac{P(AB)}{P(A)} = \frac{1}{9}$$



CS 扫描全能王

3亿人都在用的扫描App

$$(2). P(\bar{A}|B) = \frac{P(\bar{A}B)}{P(B)} = \frac{P(B) - P(AB)}{P(B)} = \frac{1}{2}$$

B3.

$$P(A) = 0.5, P(B) = 0.3, P(C) = 0.4.$$

$$P(B|A) = 0.2, P(A \cup B | C) = 0.6.$$

$$P(AB) = P(B|A) \cdot P(A) = 0.1.$$

$$P(A \cup B \cup C) = P(A \cup B) + P(C) - P((A \cup B) \cap C)$$

$$P(A \cup B) = P(A) + P(B) - P(AB) = 0.7$$

$$P((A \cup B) \cap C) = P(C) \cdot P((A \cup B) | C) = 0.24$$

$$\Rightarrow P(A \cup B \cup C) = 0.7 + 0.4 - 0.24 = 0.86$$



CS 扫描全能王

3亿人都在用的扫描App